Word Frequency Shift Across Domains

Abstract

This project analyzes how vocabulary usage varies across different domains using the Reuters-21578 dataset. By comparing word frequency distributions across domains such as corporate earnings, mergers and acquisitions, oil markets, international trade, and foreign exchange, the study demonstrates that language is highly domain-dependent. The project further identifies domain-specific vocabulary and explores a simple domain prediction approach based on word frequency patterns.

1. Introduction

Natural Language Processing (NLP) focuses on analyzing and understanding human language. A major challenge in NLP is **domain variation**, where vocabulary differs depending on the topic.

For example:

- Corporate earnings articles use words like *profit* and *revenue*
- Acquisition-related articles use *merger* and *stock*
- Oil market news uses *barrel* and *OPEC*
- Trade articles include *exports* and *tariffs*
- Currency-related news uses *dollar* and *exchange*

This project aims to analyze these differences and demonstrate how word usage shifts across domains.

2. Survey of Existing Work

Several studies in NLP have explored how vocabulary varies across domains and how this can be used for analysis.

2.1 Frequency-Based Analysis

Word frequency analysis is widely used to identify important terms in text data. Many approaches use word distributions to extract keywords and identify dominant topics.

2.2 TF-IDF and Keyword Extraction

TF-IDF assigns importance to words based on their uniqueness within documents. It is commonly used in information retrieval and document ranking.

2.3 Text Classification

Text classification methods such as Naive Bayes and Logistic Regression rely on word frequency features. These methods assume that different categories exhibit different word distributions.

2.4 Domain Variation

Previous studies show that vocabulary varies significantly across domains. Words that are frequent in one domain may be rare in another, affecting NLP model performance.

3. Dataset Description

The dataset used is the Reuters-21578 collection.

- Total documents: 21,578
- Extracted documents: 8,654
- Final filtered documents: 6,807

Each document contains:

- Topic label (domain)
- Article text

Selected Domains

- **earn** – corporate earnings
- **acq** – mergers and acquisitions
- **crude** – oil and petroleum
- **trade** – international trade
- **money-fx** – foreign exchange

4. Methodology and Implementation

The project follows this pipeline:

Dataset → Extraction → Preprocessing → Frequency Analysis → Comparison → Visualization

4.1 Data Extraction

SGML files were parsed using BeautifulSoup, and relevant fields were extracted.

4.2 Preprocessing

- Lowercasing
- Removal of punctuation
- Tokenization
- Stopword removal (including custom stopwords)

4.3 Domain-wise Corpus Construction

Documents were grouped based on domain labels.

4.4 Word Frequency Computation

Word frequencies were computed for each domain using counters.

4.5 Visualization

Bar graphs were generated to visualize domain vocabulary.

5. Results and Analysis

5.1 Top Words per Domain

Earn Domain

- profit, loss, share, quarter, revenue

Acq Domain

- shares, company, merger, acquisition, stock, agreement

Crude Domain

- oil, barrel, OPEC, production, petroleum

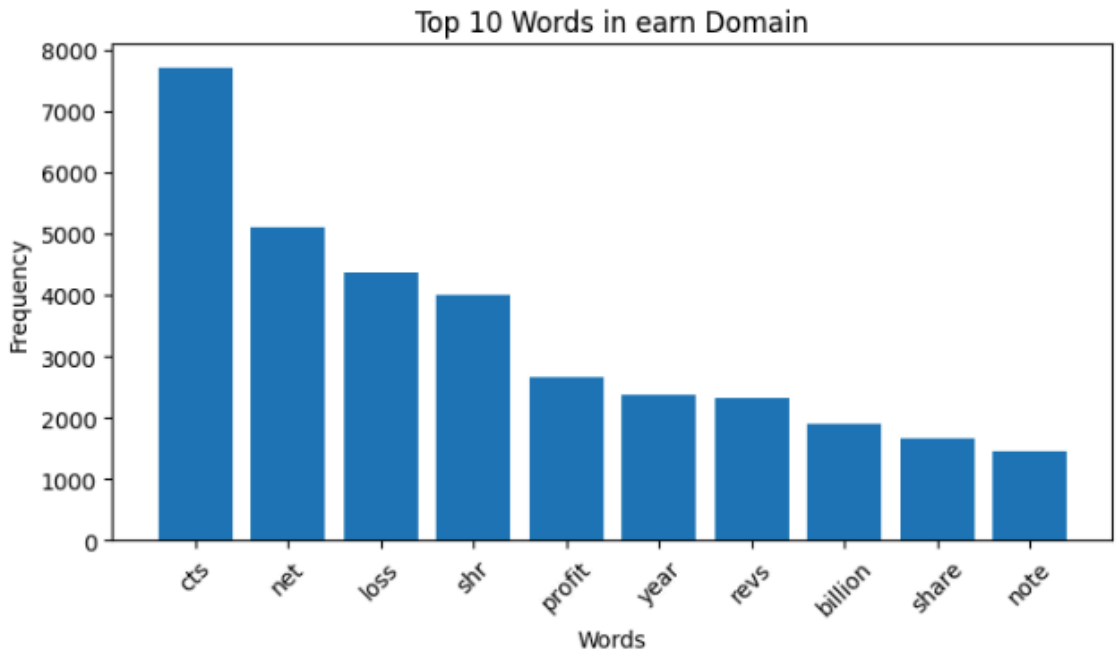
Trade Domain

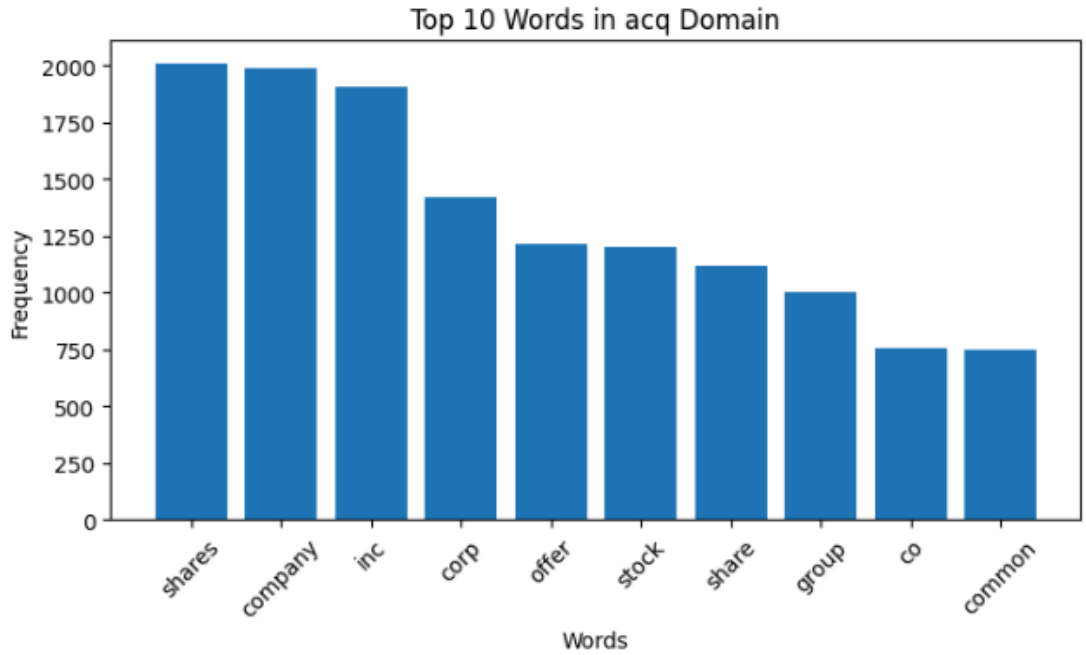
- exports, imports, tariffs, deficit

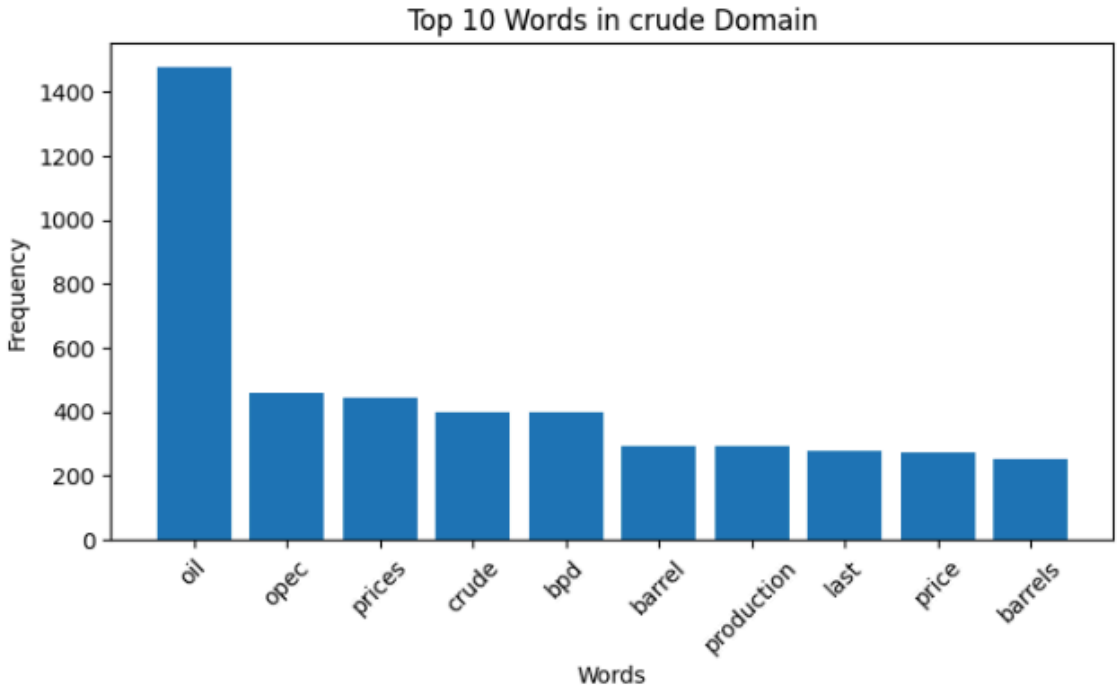
Money-FX Domain

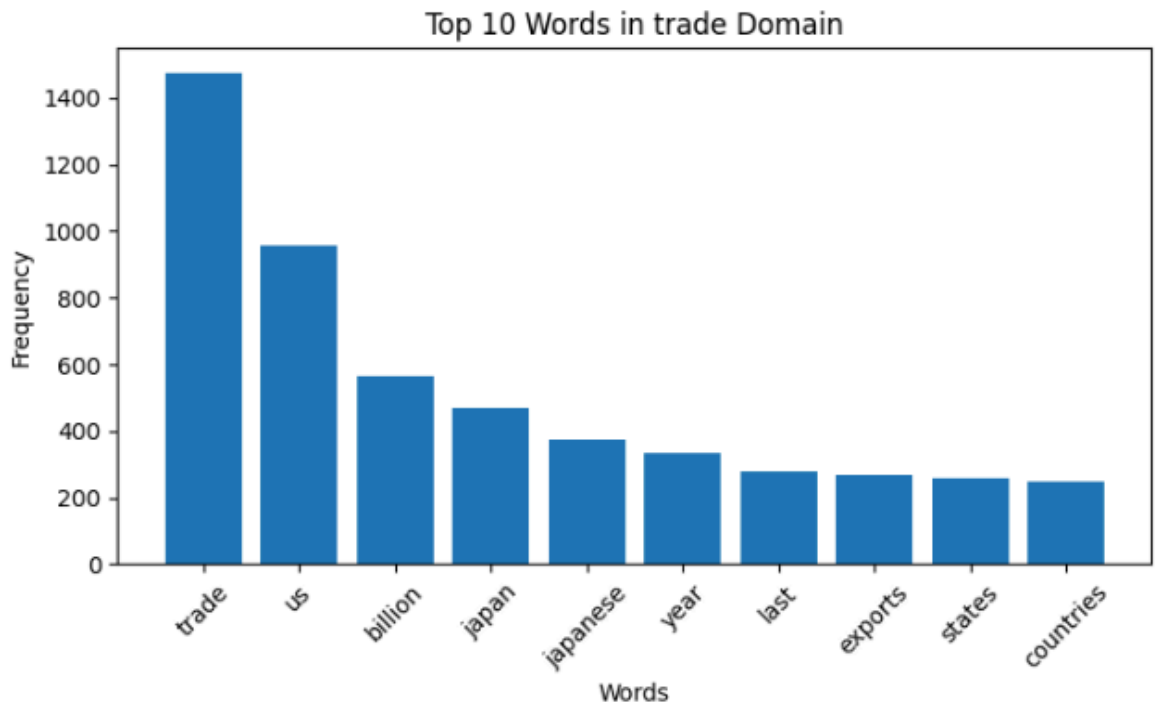
- dollar, currency, exchange, bank

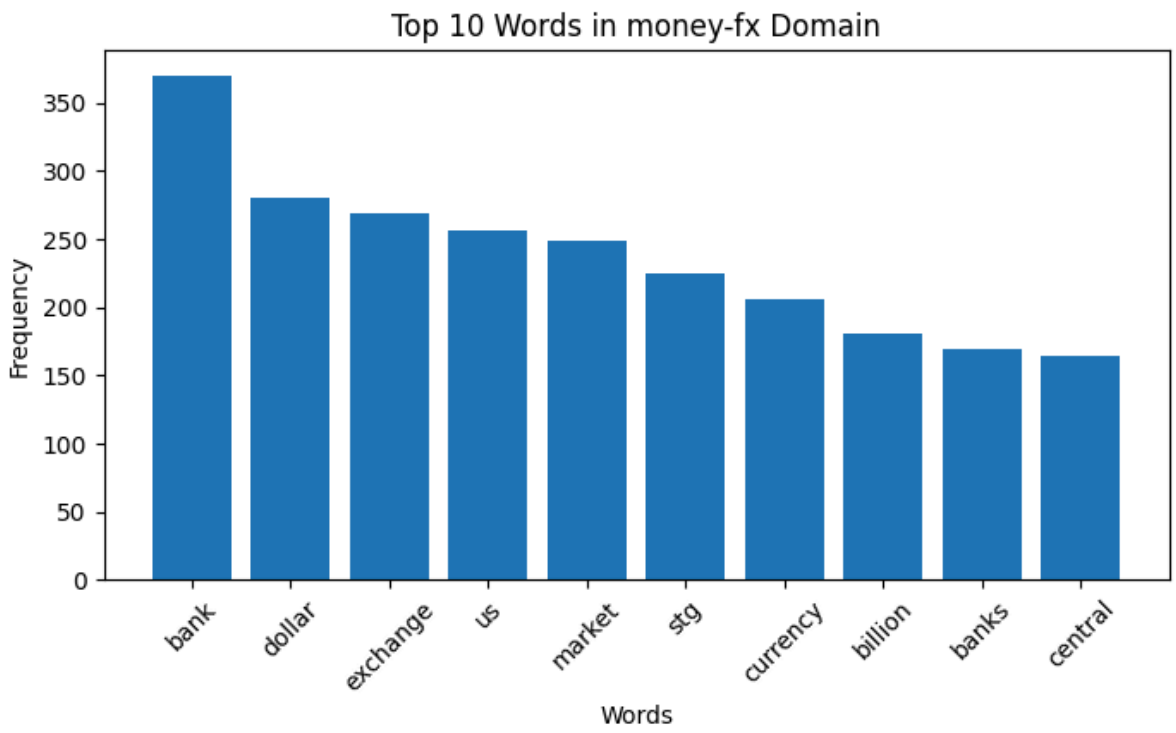
5.2 Graphical Analysis




Top 10 Words in acq Domain
Frequency
0
250
500
750
1000
1250
1500
1750
2000
shares
company
inc
corp
offer
stock
share
group
co
common
Words


Top 10 Words in crude Domain
Frequency
0
200
400
600
800
1000
1200
1400
oil
opec
prices
crude
bpd
barrel
production
last
price
barrels
Words

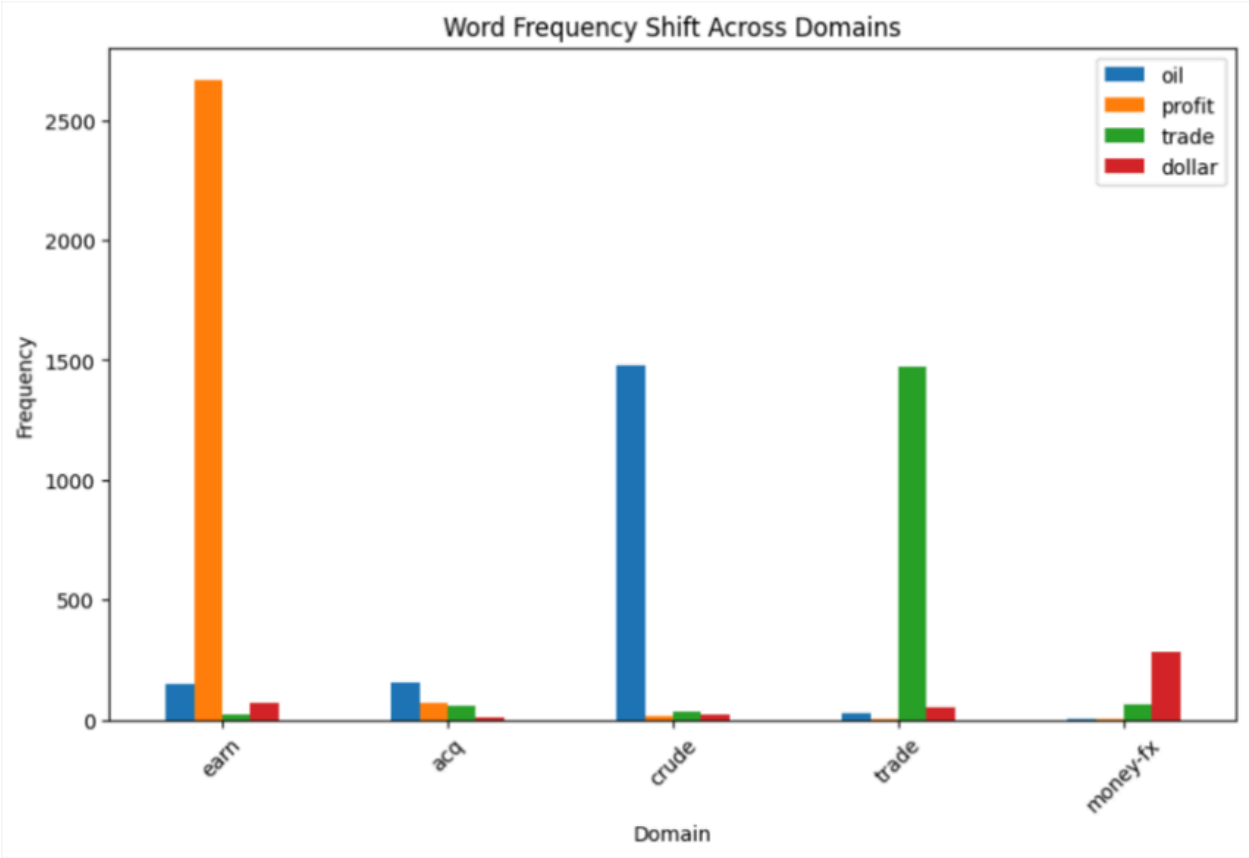




The bar charts clearly show domain-specific vocabulary patterns:

- The **earn domain** is dominated by financial performance terms such as *profit* and *quarter*
- The **acq domain** shows strong presence of *merger*, *acquisition*, and *stock*, reflecting corporate restructuring activities
- The **crude domain** contains specialized terms like *oil*, *barrel*, and *OPEC*
- The **trade domain** focuses on *exports*, *imports*, and *tariffs*
- The **money-fx domain** emphasizes *currency*, *exchange*, and *dollar*

5.3 Word Frequency Shift



Word	Dominant Domain
oil	crude
profit	earn

merger	acq
trade	trade
dollar	money-fx

This clearly demonstrates that vocabulary distribution varies across domains.

5.4 Key Observations

1. **Domain-Specific Vocabulary**
   Each domain has unique keywords.
   Example: *profit* → earn, *merger* → acq, *oil* → crude
2. **Vocabulary Overlap**
   Words like *company* and *year* appear across multiple domains.
3. **Domain Similarity**
   The *earn* and *acq* domains share vocabulary due to their financial nature.
4. **Domain Distinctiveness**
   The *crude* domain is more specialized compared to others.

6. Innovation

6.1 Distinctive Word Detection

The project identifies words that are strongly associated with specific domains based on relative frequency.

Examples:

- *merger* → acq
- *oil* → crude
- *profit* → earn

6.2 Domain Prediction Demonstration

A simple rule-based system was implemented to predict the domain of new text using vocabulary patterns.

This demonstrates how domain-specific vocabulary can be applied in real-world NLP tasks.

7. Conclusion

This project demonstrates that:

- Vocabulary varies significantly across domains
- Each domain exhibits distinct word usage patterns
- The *earn* and *acq* domains share similarities, while *crude* is more specialized
- Domain-specific vocabulary can be used for text analysis and classification

8. Future Work

- Use TF-IDF for better weighting
- Apply machine learning models
- Perform lemmatization
- Expand domain coverage